

GenMol QED Delta-y Credit Ablation

2026-09-05T21:13:25Z

Design and interpretation

Exploratory paired three-seed QED delta-y credit-assignment study using the final 50k checkpoint, a 1,000-unique-oracle-call budget, warmup parameter 100 (101 frozen iterations under the legacy iteration > warmup boundary), and neutral zero seed estimates with absolute seed scores used only for matched tie-breaking. The delta and transition-absolute control arms share parent/child oracle accounting, domains, transition deduplication, deterministic attribution, and warmup; only the post-warmup credit target differs. This QED-only reduced-budget study is not paper-comparable, causal, or confirmatory.

The primary paired contrast is delta minus running_mean_delta_control. Both arms freeze vocabulary updates during warmup, score the same parent/child domains, deduplicate canonical parent-child transitions, and use the same deterministic changed-fragment mapping. Both initialize online estimates at zero and use absolute seed scores only to break ties. Their sole configured difference is post-warmup credit: child minus parent versus absolute child. Released and ordinary running mean are contextual references only.

Exact metric definitions

Quantity	Definition
Final top-k	At a stated horizon, top-k is the arithmetic mean of the highest min(k, N) QED scores observed in that view, where N is the number of included scores available by that horizon.
AUC top-k	The top-k trajectory starts at (0, 0), is evaluated every 100 axis units through the stated horizon, is integrated by the trapezoidal rule, and is divided by that horizon.

Observed paired effect (not a claim of efficacy)

Metric	Mean delta-control difference	Seed 0	Seed 1	Seed 2
auc_top_1	-0.000615 +/- 0.001224	+0.000179	-0.002024	+0.000000
auc_top_10	-0.001309 +/- 0.010332	+0.009665	-0.010849	-0.002742
auc_top_100	-0.005747 +/- 0.012864	+0.007513	-0.018175	-0.006580
top_1	-0.000202 +/- 0.000834	+0.000512	-0.001118	+0.000000
top_10	-0.001428 +/- 0.013455	+0.014069	-0.010130	-0.008224
top_100	-0.014601 +/- 0.017650	+0.005667	-0.022885	-0.026585

Positive values favor delta. Sample SD describes three paired seeds; no confidence interval or significance decision is justified here.

Aggregate metric views

All-call metrics

Every unique parent and child oracle evaluation occupies its true position on the 1,000-call axis. Values are mean +/- sample SD across seeds 0, 1, and 2.

Variant	AUC top-1	AUC top-10	AUC top-100	Final top-1	Final top-10	Final top-100
released (contextual)	0.899727 +/- 0.000983	0.894028 +/- 0.001708	0.853075 +/- 0.003966	0.948252 +/- 0.000240	0.946821 +/- 0.000549	0.933910 +/- 0.003292
running_mean (contextual)	0.899855 +/- 0.001424	0.895869 +/- 0.002388	0.857370 +/- 0.004860	0.948149 +/- 0.000316	0.947197 +/- 0.000532	0.938896 +/- 0.001739
transition-absolute control	0.897520 +/- 0.002617	0.880752 +/- 0.008178	0.826609 +/- 0.013075	0.945197 +/- 0.003324	0.933299 +/- 0.007924	0.903087 +/- 0.011652
delta-y	0.896905 +/- 0.002069	0.879443 +/- 0.005787	0.820861 +/- 0.010052	0.944994 +/- 0.002923	0.931870 +/- 0.009452	0.888486 +/- 0.015752

Charged-child score subset on the total unique-call axis (not equal child evaluations)

Variant	Children	AUC top-1	AUC top-10	AUC top-100	Final top-1	Final top-10	Final top-100
released (contextual)	1000.0 +/- 0.0	0.899727 +/- 0.000983	0.894028 +/- 0.001708	0.853075 +/- 0.003966	0.948252 +/- 0.000240	0.946821 +/- 0.000549	0.933910 +/- 0.003292
running_mean (contextual)	1000.0 +/- 0.0	0.899855 +/- 0.001424	0.895869 +/- 0.002388	0.857370 +/- 0.004860	0.948149 +/- 0.000316	0.947197 +/- 0.000532	0.938896 +/- 0.001739
transition-absolute control	534.3 +/- 4.5	0.896374 +/- 0.001905	0.875034 +/- 0.004474	0.808135 +/- 0.010234	0.943835 +/- 0.002018	0.927621 +/- 0.005753	0.884322 +/- 0.009591
delta-y	533.7 +/- 3.2	0.895007 +/- 0.003177	0.870624 +/- 0.006470	0.799716 +/- 0.011986	0.942372 +/- 0.002995	0.922702 +/- 0.008617	0.866259 +/- 0.015298

Charged-child scores retain their actual global call indices; parent calls create gaps. The count column is the number of unique charged-child evaluations reached, not the number of runner iterations.

Common first 500 charged child evaluations

Variant	AUC top-1	AUC top-10	AUC top-100	Final top-1	Final top-10	Final top-100
released (contextual)	0.851326 +/- 0.001593	0.841772 +/- 0.002973	0.781009 +/- 0.004145	0.947803 +/- 0.000560	0.945506 +/- 0.000622	0.913409 +/- 0.005949
running_mean (contextual)	0.851570 +/- 0.002543	0.844883 +/- 0.003926	0.781868 +/- 0.008355	0.948056 +/- 0.000273	0.946174 +/- 0.001459	0.922238 +/- 0.003357
transition-absolute control	0.848912 +/- 0.001915	0.827786 +/- 0.003649	0.758393 +/- 0.009535	0.943835 +/- 0.002018	0.927196 +/- 0.005456	0.881286 +/- 0.010538
delta-y	0.847889 +/- 0.003028	0.823719 +/- 0.005778	0.751477 +/- 0.010315	0.942372 +/- 0.002995	0.921793 +/- 0.008515	0.864572 +/- 0.015623

The first 500 charged child QED scores in event order, densely indexed as charged-child evaluations 1 through 500; checkpoints are 0, 100, ..., 500 and the trapezoidal AUC divisor is 500 charged-child evaluations.

Delta-y and attribution diagnostics

Diagnostics are recomputed from hash-checked event streams. Delta distributions and both attribution summaries use unique canonical parent-child transitions, matching the estimator's deduplication unit. Fragment coverage is the credited child-fragment fraction, averaged across those unique transitions.

Variant / seed	Charged parents	Cached parents / all lookups	Charged children	Unique transitions	Delta-y mean +/- SD	+ / 0 / -	Transitions with any credit	Mean fragment coverage	Credited observations / unique
transition-absolute control / 0 events 604; cached child 73	470	33/503 (0.066)	530	501	-0.01213 +/- 0.13615	180 / 66 / 255	437/501 (0.872)	0.532	5087 / 4458
transition-absolute control / 1 events 600; cached child 66	466	33/499 (0.066)	534	499	-0.03265 +/- 0.10237	141 / 58 / 300	443/499 (0.888)	0.539	3810 / 3124
transition-absolute control / 2 events 586; cached child 46	461	24/485 (0.049)	539	484	-0.02183 +/- 0.12021	185 / 38 / 261	446/484 (0.921)	0.557	4681 / 3966
delta-y / 0 events 607; cached child 70	464	42/506 (0.083)	536	505	-0.03884 +/- 0.11903	160 / 57 / 288	444/505 (0.879)	0.528	4235 / 3620
delta-y / 1 events 582; cached child 47	465	16/481 (0.033)	535	481	-0.02245 +/- 0.13717	170 / 41 / 270	441/481 (0.917)	0.550	4118 / 3525
delta-y / 2 events 586; cached child 55	470	15/485 (0.031)	530	483	+0.01521 +/- 0.14107	218 / 38 / 227	446/483 (0.923)	0.557	5497 / 4696

Matched charged-child subset on total unique-call axis (not equal child evaluations)

Metric	Delta minus control	Seed deltas
auc_top_1	-0.001366 +/- 0.002367	s0=+0.000000, s1=-0.004099, s2=+0.000000
auc_top_10	-0.004411 +/- 0.009931	s0=+0.006771, s1=-0.012207, s2=-0.007796
auc_top_100	-0.008419 +/- 0.014433	s0=+0.007324, s1=-0.021028, s2=-0.011554
top_1	-0.001462 +/- 0.002533	s0=+0.000000, s1=-0.004387, s2=+0.000000
top_10	-0.004919 +/- 0.012346	s0=+0.008886, s1=-0.008740, s2=-0.014903
top_100	-0.018064 +/- 0.018292	s0=+0.003054, s1=-0.029021, s2=-0.028223

Matched first-500-charged-child diagnostic

Metric	Delta minus control	Seed deltas
auc_top_1	-0.001024 +/- 0.001773	s0=+0.000000, s1=-0.003071, s2=+0.000000
auc_top_10	-0.004067 +/- 0.008871	s0=+0.005800, s1=-0.011381, s2=-0.006621
auc_top_100	-0.006916 +/- 0.011902	s0=+0.005755, s1=-0.017860, s2=-0.008643
top_1	-0.001462 +/- 0.002533	s0=+0.000000, s1=-0.004387, s2=+0.000000
top_10	-0.005402 +/- 0.012481	s0=+0.008886, s1=-0.010912, s2=-0.014180
top_100	-0.016714 +/- 0.018862	s0=+0.004921, s1=-0.029702, s2=-0.025361

This equalizes 500 unique charged-child evaluations per run. Its AUC uses dense charged-child indices 1..500, the origin (0, 0), 100-evaluation checkpoints, trapezoids, and a divisor of 500; it does not equalize total oracle calls or runner iterations.

Configuration and provenance

Item	Value
Model checkpoint	50000.ckpt SHA-256 8d00aa47b02f64bf39ff6b0b2e786f213587366fc2c3d29712a00f3f84108dd6
Initial QED vocabulary	SHA-256 6b98420f3fa88835e50cb28b960225dc6c7857a29f3057fb1ca22b4211c1806c
Matrix	qed_50k_delta_1k_v2.yaml SHA-256 8c62dd6c040392be48a1d6f1024179e16b2b70ff2667073658affd4d89197dc8
Collection	collection_manifest.json SHA-256 0f4d19d3a1e01ad99346c05415343d28ef95e54418fd89baa5c80e25b6f5ea5d
Results CSV	results.7dde67317ac09075fc0510b94fe2a7635ad7ed00f17595f0b92f854caefedda7.csv SHA-256 7dde67317ac09075fc0510b94fe2a7635ad7ed00f17595f0b92f854caefedda7
Reporter	/home/aidar.alimbayev/Documents/genmolv2/scripts/exps/pmo/report_delta_ablation.py SHA-256 ec5ac38463a34fb9edaba60fec0fd7535fde29df4c4605863f0186a861bb5072
Code identity	7f401b8217e86afce7b725c7a6a25480d230214b dirty=False tracked diff=e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
GPU sharing	All 12 jobs shared a GPU; wall-time comparisons suppressed
GPU launches	GPU 5 GPU-134e9a1b-cf80-5c0e-b0c4-048fff8e6d3c: 3 jobs; launch utilization 5-5%; minimum free 47329 MiB GPU 7 GPU-7ae42da2-3c8d-efa0-99ec-845b7f85399a: 3 jobs; launch utilization 4-6%; minimum free 47161 MiB GPU 2 GPU-997e881a-bd08-aa54-45f3-9b3c6dfd6ece: 3 jobs; launch utilization 5-5%; minimum free 47165 MiB GPU 6 GPU-d7e40ce5-9260-a208-16a0-3372e3f204de: 3 jobs; launch utilization 5-5%; minimum free 47331 MiB

Locked hyperparameters

Budget	Warmup	Population	gamma	Temperature	Randomness	Guidance	Attribution
1,000 unique calls	parameter 100; 101 frozen (0..100)	100	0.0	1.2	2.0	2.0	novel_vs_parent

Estimator controls

Arm	Seed initialization	Post-warmup credit value
released (contextual)	released_absolute_rows	absolute_child_score
running_mean (contextual)	absolute_seed_statistics	absolute_child_score
transition-absolute control	neutral_zero_with_seed_score_tiebreak	absolute_child_score
delta-y	neutral_zero_with_seed_score_tiebreak	child_score_minus_parent_score

Scientific caveats

- This is an exploratory QED-only, 1,000-call, three-seed experiment; it is neither confirmatory nor paper-comparable.
- Delta-y is an association for an adaptive parent-child proposal, not a causal fragment contribution.
- The deterministic all-cut child-minus-parent mapping is approximate; several credited fragments each receive the full molecular delta.
- Parent-scoring arms spend part of the fixed budget on parents and therefore evaluate fewer children than released or ordinary running mean.
- Both matched arms start with neutral zero online estimates; absolute seed scores only break ties. The released and ordinary running-mean arms are contextual references, not matched causal controls.
- A unique oracle call is a newly scored canonical molecule. A charged child is a unique child evaluation, not an optimizer iteration; logged events and cached-child lookups are reported separately. Cached parent lookups are free under the unique-call budget.
- The first-500-child view equalizes charged child evaluations, but those children can occur at different total oracle-call positions and remain adaptively generated.
- All 12 jobs shared a GPU. Score comparisons remain valid, but wall-time comparisons are suppressed.